UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ASCEND 2030

SCST1133 Data Engineering Ecosystem
Semester 1 · Session 2026/2027

Topic 3: Introduction to Data Engineering

Assoc. Prof. Dr. Mohd Shahizan Othman

Faculty of Computing · Universiti Teknologi Malaysia

Scan for the Topic 3 web page

Innovating Sustainable Solutions

Warm-up · think, then share

1

Name one app you used today that runs on data.

2

Who prepared that data before it reached your screen?

Share with the person next to you, then post one answer in the class chat.

Learning outcomes · CLO1

By the end of today, you can…

- Explain what data engineering is and what a data engineer does
- Trace the evolution of the field across four eras
- Distinguish data engineering from data science
- Identify the five lifecycle stages and six undercurrents
- Describe core skills, trade-offs and primary languages
- Discuss trends shaping the future of the field

CLO1: Explain basic concepts of information technology and their role in supporting data-driven systems through systems thinking.

SCST1133 course information; Topic 3 web module.

Course map

Where Topic 3 fits



- Topic 3 draws the map of the whole discipline
- Topics 4, 9, 10, 12 and 13 each zoom into one stage or undercurrent
- Topic 5 adds the people and organisations around the data

Source: SCST1133 course topics, 2026/2027

Today · three hours

Four parts, one break

01 Part 1 · Define and trace the field

02 Part 2 · Distinguish and map the lifecycle

03 Break · 10 minutes

04 Part 3 · Skills, toolkit and the future

05 Part 4 · Case studies and the graded task

06 Wrap-up · takeaways and next steps



drshahizan.github.io/scst1133/topic-03

Notes, activities, quiz and case studies



UTM
UNIVERSITI TEKNOLOGI MALAYSIA
ASCEND 2030

Define and trace the field

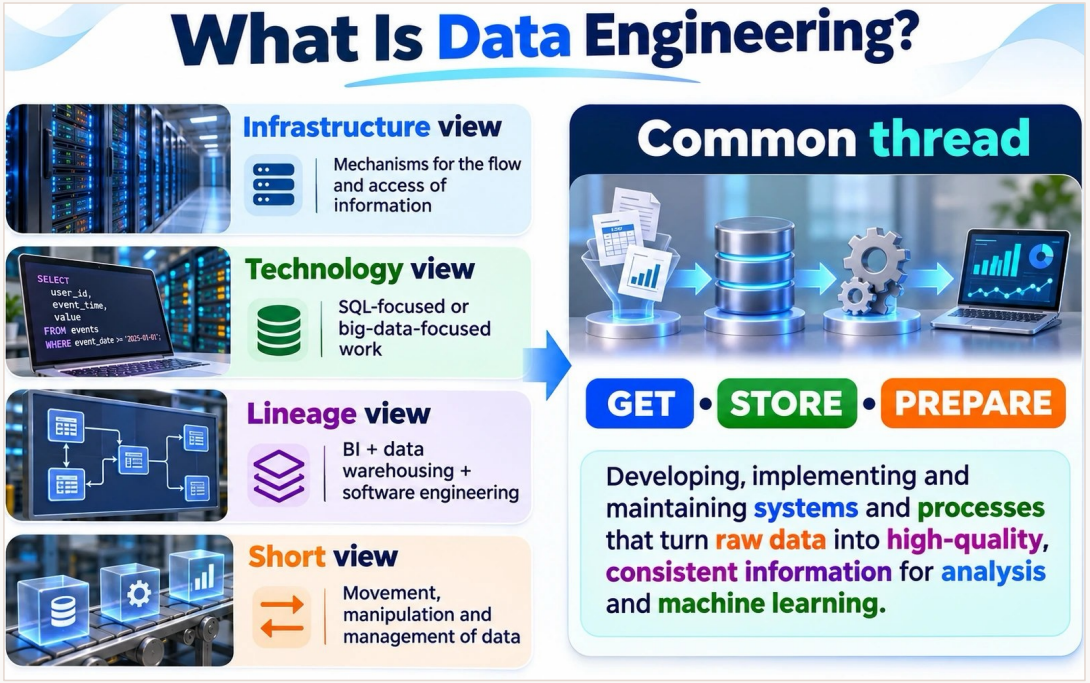
Part 1

Steps 01–02 · about 45 minutes

Definition · evolution · big data · privacy

Step 01 · Define

Many definitions, one common thread



- Experts describe the field from very different angles
- Common thread: get, store and prepare data for others to use
- Data engineering is where six undercurrents meet

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, pp. 3–5

Step 01 · Define

A working definition

Data engineering is developing, implementing and maintaining the systems and processes that turn raw data into high-quality, consistent information for downstream uses such as analysis and machine learning.

A data engineer manages the data engineering lifecycle—from getting data out of source systems to serving it for use.

Paraphrased from Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, p. 4.

Activity 1 · individual · 5 minutes

Definition remix

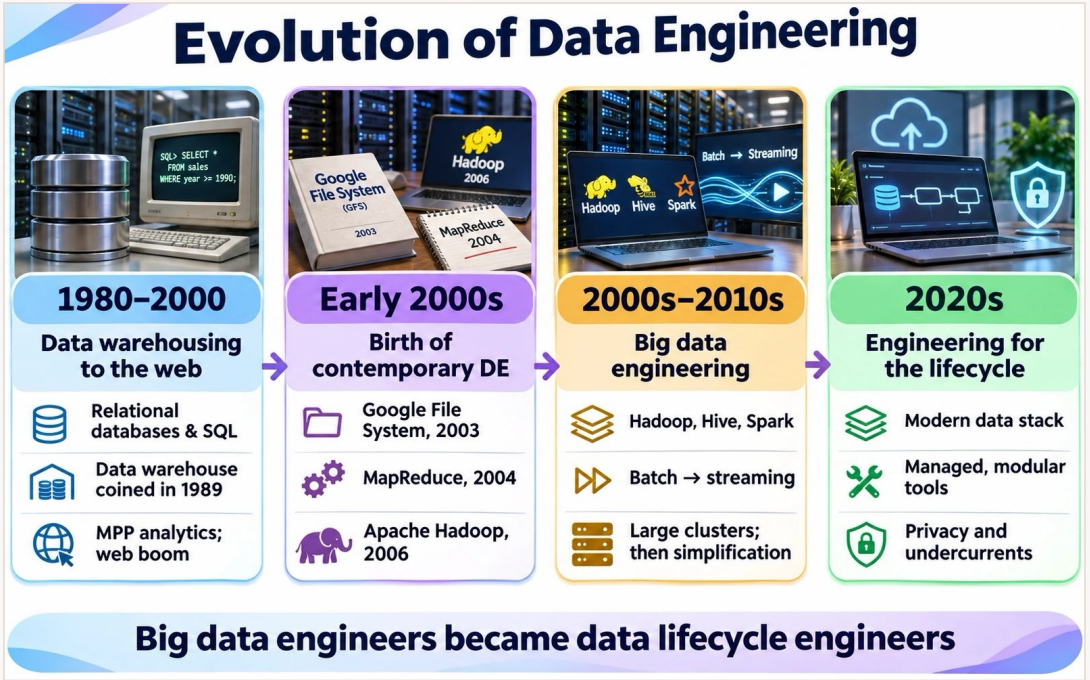
Write your own definition of data engineering in one sentence.

- Use no more than 30 words
- Mention where data comes from
- Mention who uses the result
- Swap with a partner: what did each of you include that the other missed?

Topic 3 web module, Activity 1.

Step 02 · Trace

Four eras of data engineering



- 1980–2000: relational databases, SQL and data warehouses
- Early 2000s: GFS, MapReduce, Hadoop and AWS
- 2000s–2010s: big data engineering, then simplification
- 2020s: the data lifecycle engineer

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, pp. 6–11

2003

Google File System: store huge files on many cheap machines

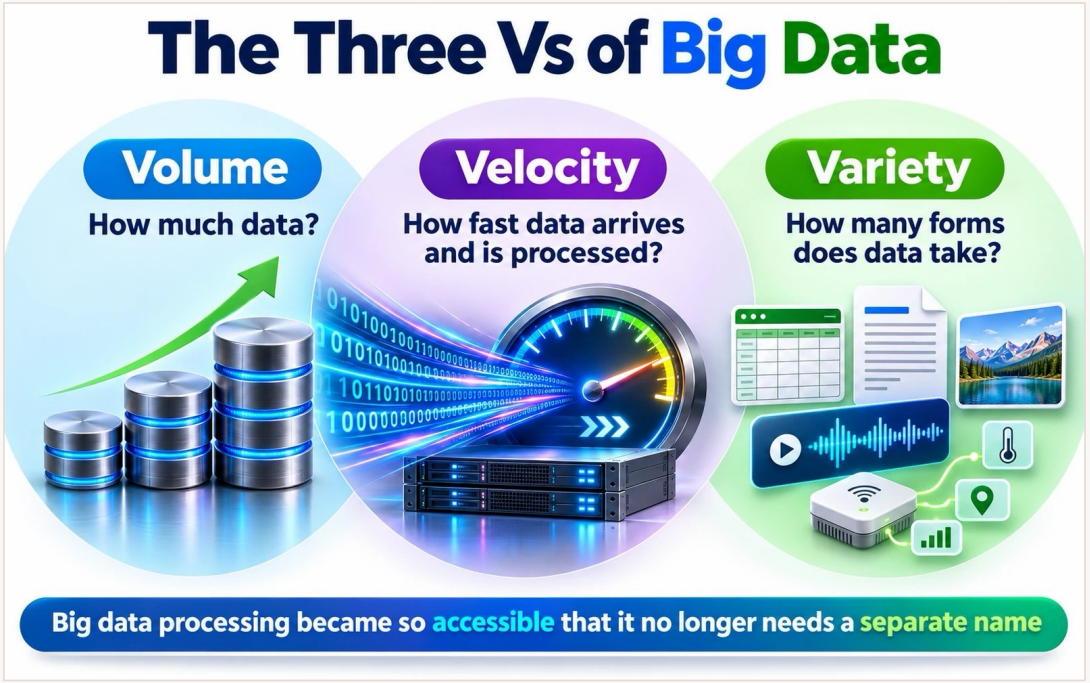
2004

MapReduce: split huge jobs into parallel steps

Yahoo engineers open-sourced Hadoop in 2006, and Amazon Web Services made computing rentable.

Step 02 · Critical thinking

Big data: hype and reality



- Excitement: big-company tools for everyone
- Hype: Hadoop clusters for a few gigabytes
- Cost: teams spent more time maintaining than delivering
- Simplification: big data engineers became simply data engineers

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, pp. 7–9

2010

Personal Data Protection Act passed in Malaysia

2024

Amendment Act; phased into force during 2025

New duties: breach notification and data protection officers; fines up to RM1 million.

Quick check 1 of 3 · vote with your fingers

Why did the term “big data” fade?

A Data stopped growing

B Large-scale processing became simple and accessible

C Governments banned big data tools

D Spreadsheets replaced databases

Answer on the next slide.

Quick check 1 · answer

Why did the term “big data” fade?

A Data stopped growing

B Large-scale processing became simple and accessible ✓

C Governments banned big data tools

D Spreadsheets replaced databases

Managed, simpler tools made large-scale processing ordinary—so big data engineers became simply data engineers.

Topic 3 web module, Knowledge Check.



UTM UNIVERSITI TEKNOLOGI MALAYSIA

ASCEND 2030

Distinguish and map the lifecycle

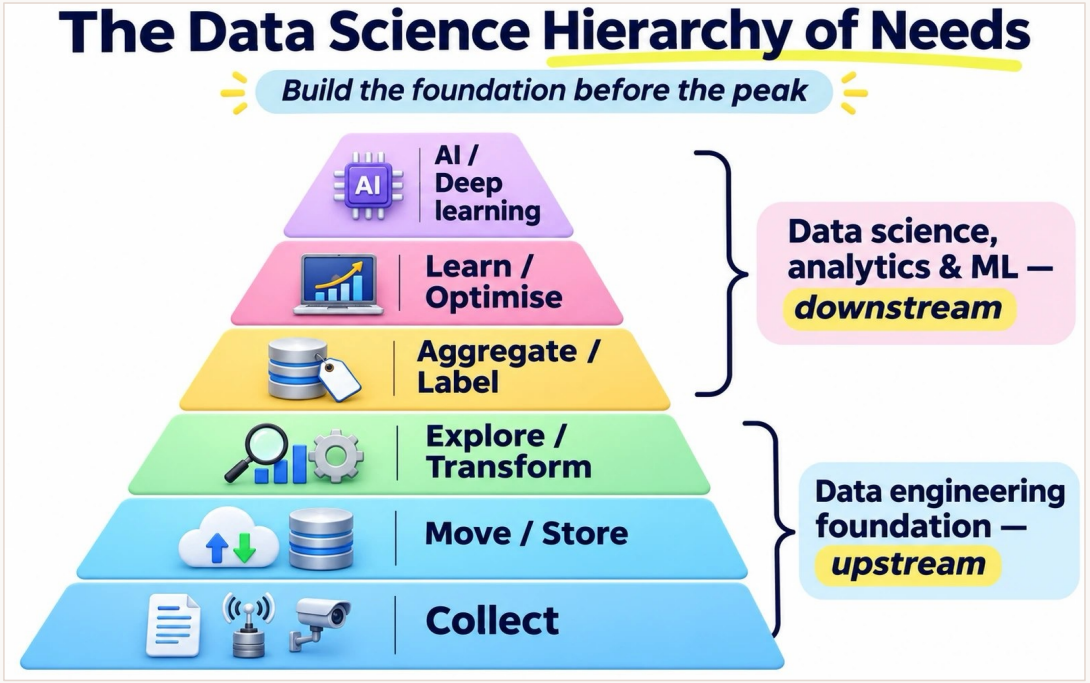
Part 2

Steps 03–05 · about 50 minutes

DE vs DS · lifecycle · undercurrents · PHE case

Step 03 · Distinguish

Data engineering sits upstream



70–80%

of data scientists' time is spent gathering, cleaning and processing data

- Data engineers build the lower levels of the pyramid
- Then analytics, experiments and AI can succeed

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, pp. 11–12; Rogati (2017)

Activity 2 · pairs · 10 minutes

Hierarchy triage: DE or DS?

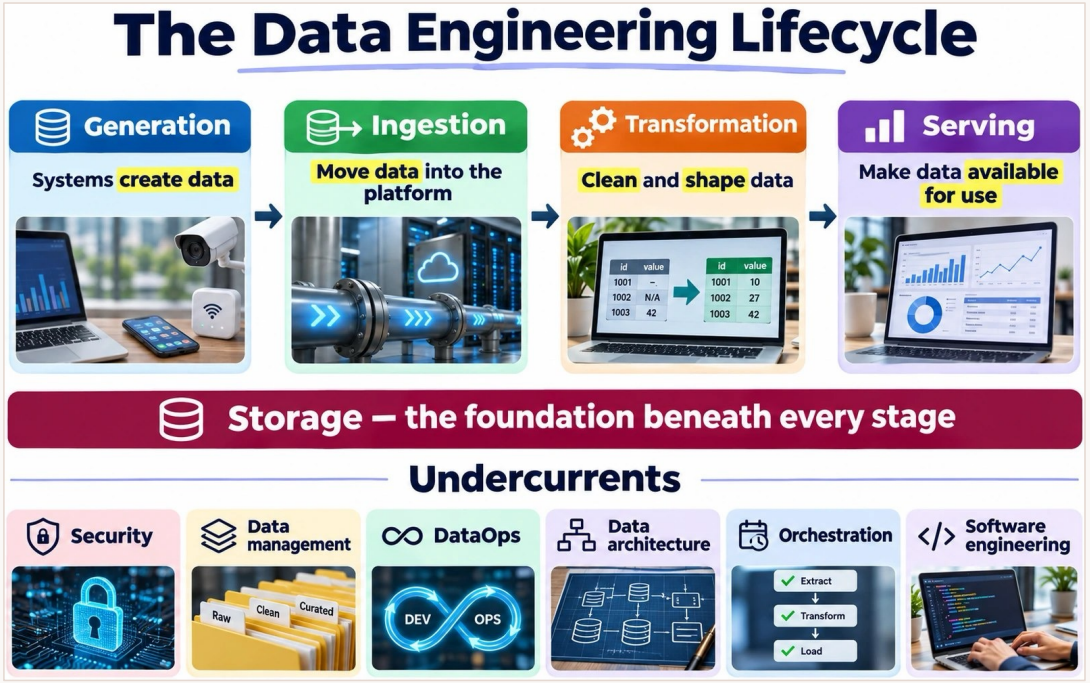
Place each task on the pyramid, then label it DE or DS.

- Set up logging so every app click is recorded
- Train a model that predicts which students may drop a course
- Copy factory sensor readings into central storage every minute
- Run an A/B test of two website layouts
- Convert dates from many formats into one standard
- Schedule a nightly job and alert the team if it fails

Topic 3 web module, Activity 3 (full list of 12 tasks online).

Step 04 · Map

Five stages from source to use

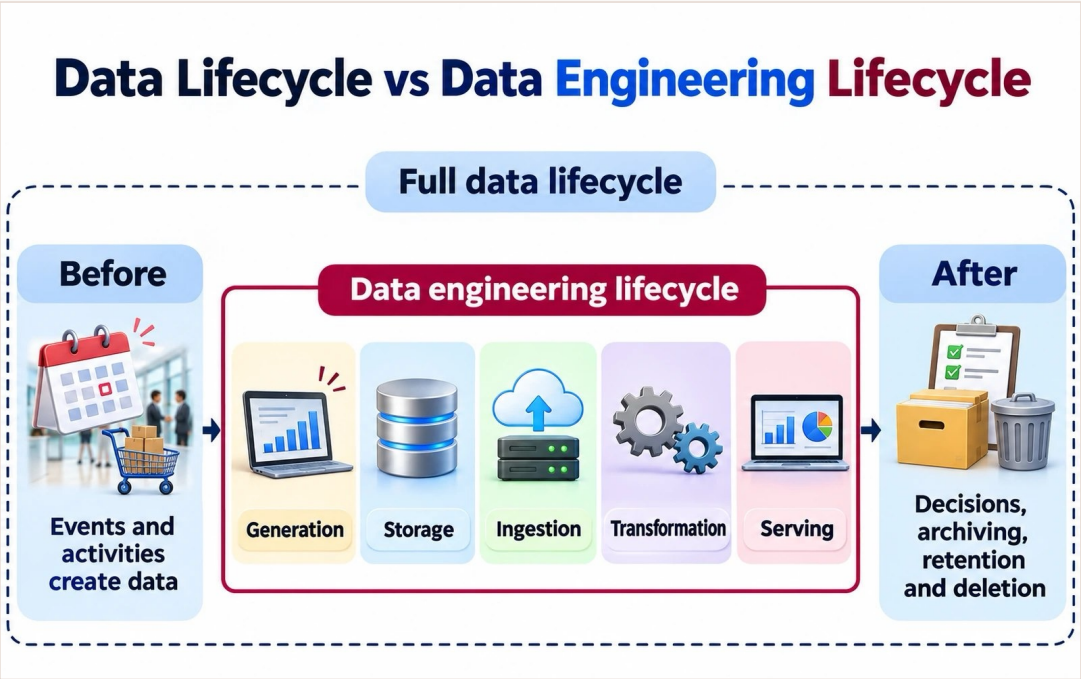


- Generation: where data comes from
- Storage: the foundation beneath every stage
- Ingestion: moving data in
- Transformation: making it useful
- Serving: analytics, ML and apps

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, p. 5; Ch. 2, pp. 33–35

Step 04 · Map

The engineering lifecycle is a subset

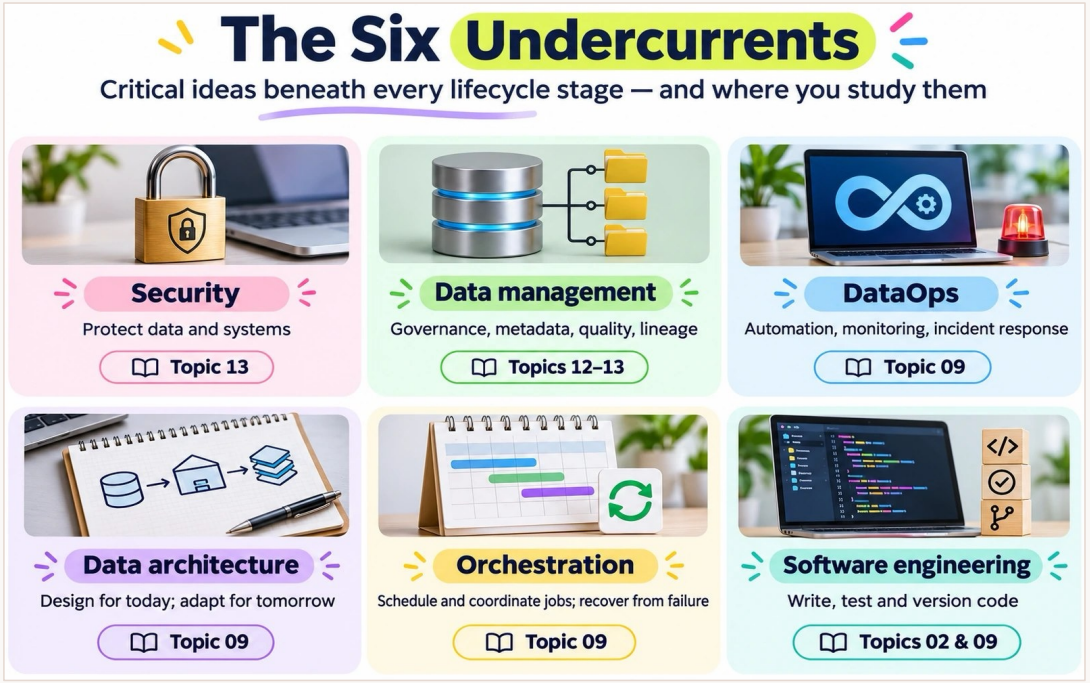


- Full data lifecycle: the entire lifespan of data
- Data engineering lifecycle: the stages an engineer controls
- Topic 5 follows the organisational journey; Topic 9 builds the engineering lifecycle

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 2, p. 34

Step 05 · Map

Six undercurrents run beneath every stage

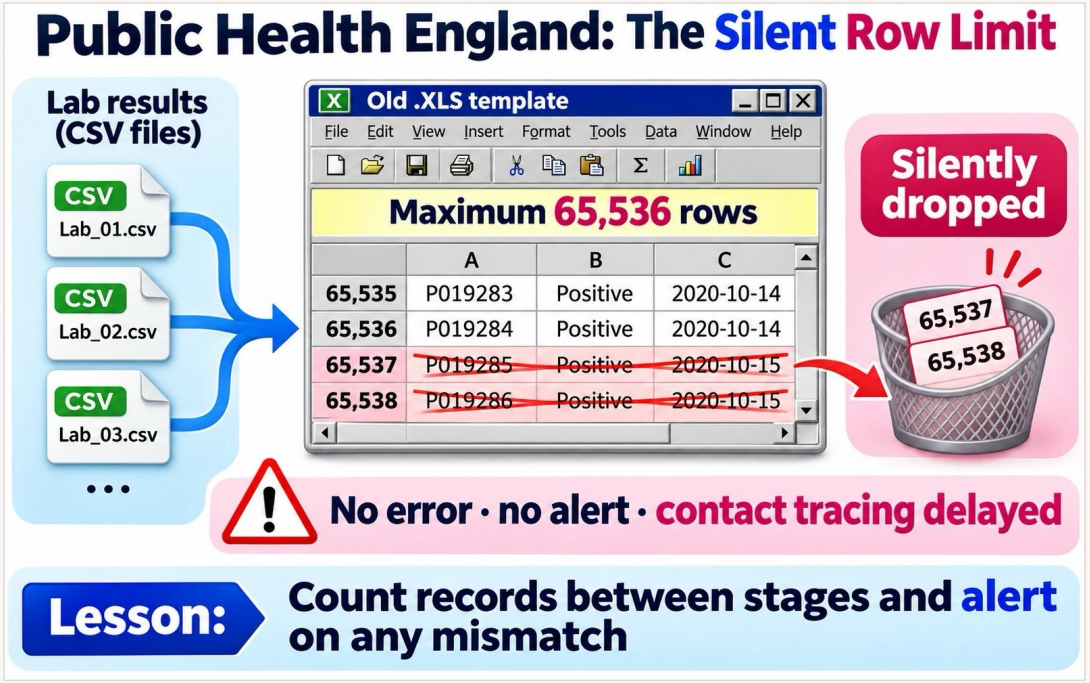


- Like currents under the sea: unseen, yet they steer everything
- A perfect pipeline still fails if it is insecure or unmonitored
- Using Git in Topic 2 was software engineering in action

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 2, pp. 48–68

Mini case · when undercurrents fail

15,841 cases went missing



- England, Sept–Oct 2020: lab results loaded into an old .XLS template
- Each sheet holds 65,536 rows; extra records were silently dropped
- Contact tracing was delayed
- A record-count check and an alert would have caught it

Source: The Register (2020); Topic 3 optional case

Quick check 2 of 3 · vote with your fingers

Which stage underpins all the others?

A Generation

B Storage

C Serving

D Ingestion

Answer on the next slide.

Quick check 2 · answer

Which stage underpins all the others?

A Generation

B Storage ✓

C Serving

D Ingestion

Data is stored at many points as it flows, so storage is drawn as the foundation of the lifecycle.

Topic 3 web module, Knowledge Check.

Activity 3 · groups · 20 minutes

Label the lifecycle

A smart electricity meter sends a reading every 30 minutes. The owner sees a daily chart in an app, gets an alert when usage spikes and receives a monthly bill.

- Label the five stages: generation, storage, ingestion, transformation, serving
- For each undercurrent, describe one thing that could go wrong
- Which stage is most critical for this service? Justify it

Topic 3 web module, Activity 4.



Stretch, refill,
return in 10 minutes

Break

Part 3 starts at the next slide

Scan the QR code on slide 1 if you missed anything



UTM
UNIVERSITI TEKNOLOGI MALAYSIA
ASCEND 2030

Skills, toolkit
and the future

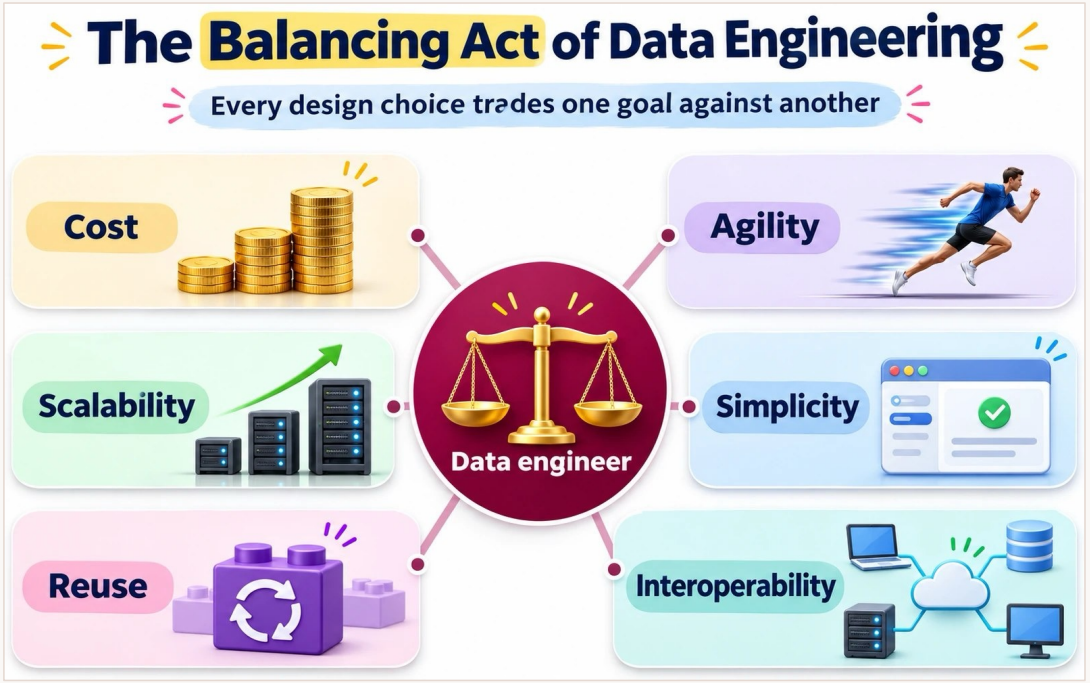
Part 3

Steps 06–08 · about 40 minutes

Trade-offs · languages · SQL · trends

Step 06 · Describe

Every design choice is a trade-off



- Balance cost, agility, scalability, simplicity, reuse and interoperability
- Data engineers usually do not build ML models, dashboards or KPIs
- …but must understand them to serve the people who do

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, p. 13

Activity 4 · two teams · 15 minutes

Balancing-act debate

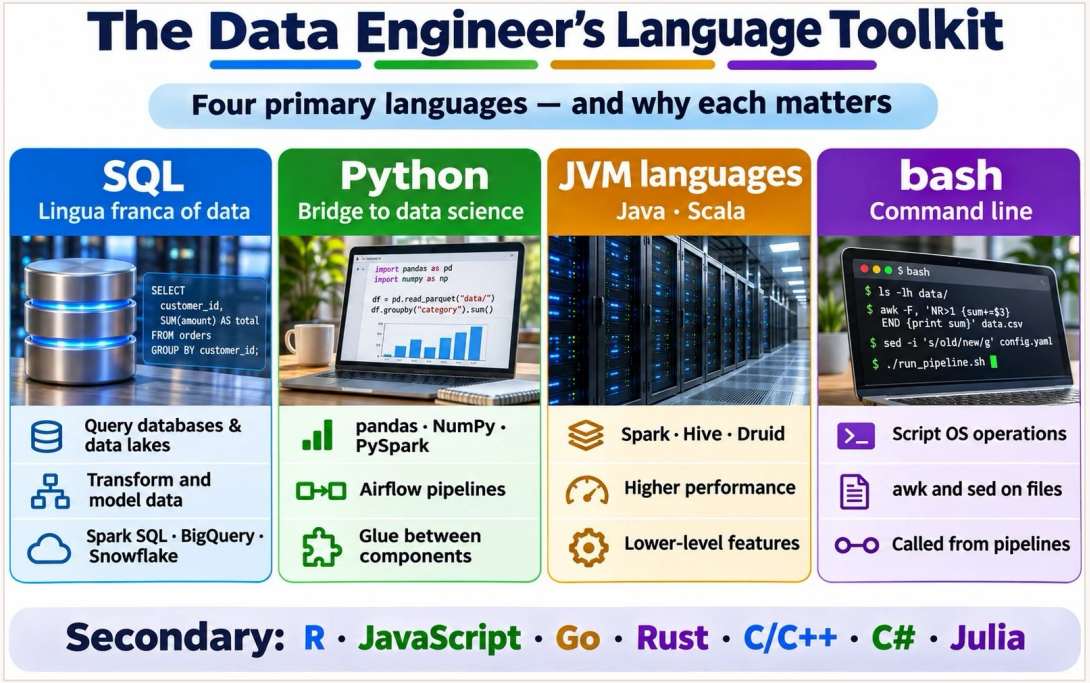
Team A is a five-person start-up launching a food-review app. Team B is a national bank.

- Rank the six goals from most to least important for your organisation
- Defend your top two and bottom two in two minutes
- Then compare: why do the rankings differ?

Topic 3 web module, Activity 5.

Step 07 · Describe

Four primary languages

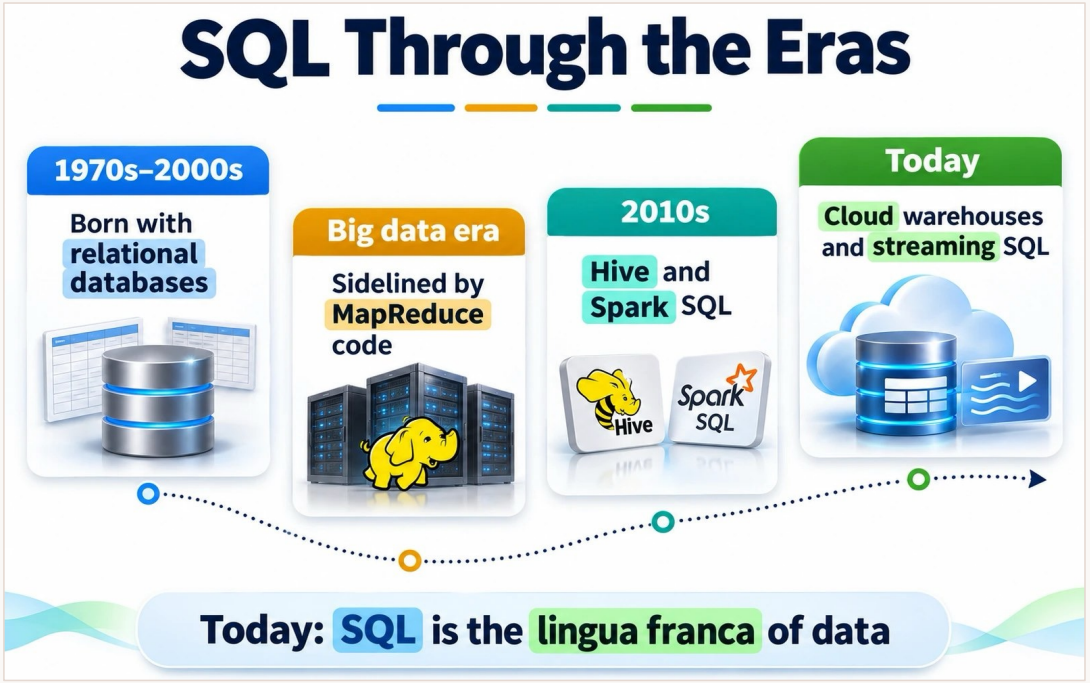


- SQL: the lingua franca of data—aim to be highly proficient
- Python: the bridge to data science
- Focus on fundamentals; watch where the field is going

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, pp. 19–21

Step 07 · What's old is new again

The comeback of SQL

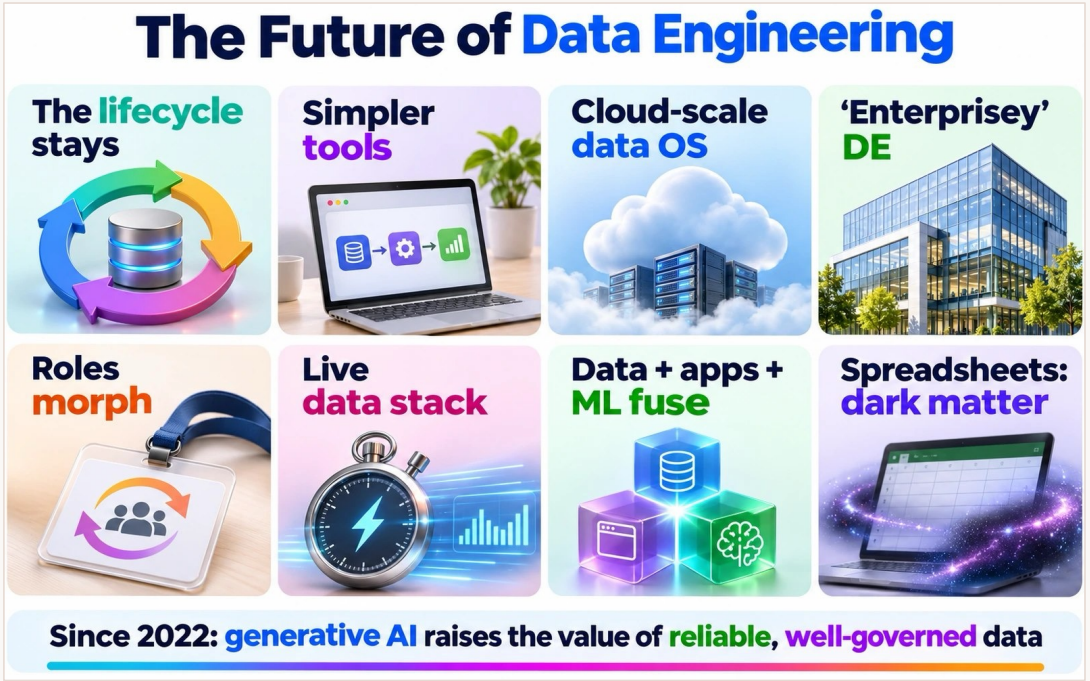


- Born with relational databases in the 1970s
- Sidelined by custom MapReduce code
- Back through Hive, Spark SQL, cloud warehouses and streaming SQL
- Declarative: describe what you want, not how

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 1, pp. 20–21

Step 08 · Anticipate

Tools change; fundamentals remain



- The lifecycle is not going away
- Simpler tools, live data, and data fused with apps and ML
- Since 2022, generative AI makes reliable, governed data even more valuable

Source: Reis & Housley (2022), Fundamentals of Data Engineering, Ch. 11, pp. 379–389

Quick check 3 of 3 · vote with your fingers

What does a data engineer usually NOT do?

A Build a data pipeline

B Create a business dashboard

C Monitor data quality

D Secure data access

Answer on the next slide.

Quick check 3 · answer

What does a data engineer usually NOT do?

A Build a data pipeline

B Create a business dashboard ✓

C Monitor data quality

D Secure data access

Dashboards are usually built by analysts—but data engineers must understand them to serve their users.

Topic 3 web module, Knowledge Check.



UTM UNIVERSITI TEKNOLOGI MALAYSIA

ASCEND 2030

Case studies and
the graded task

Part 4

Seven real stories · about 25 minutes

Uber · KKM · five optional cases

Case studies

Seven real stories, two required

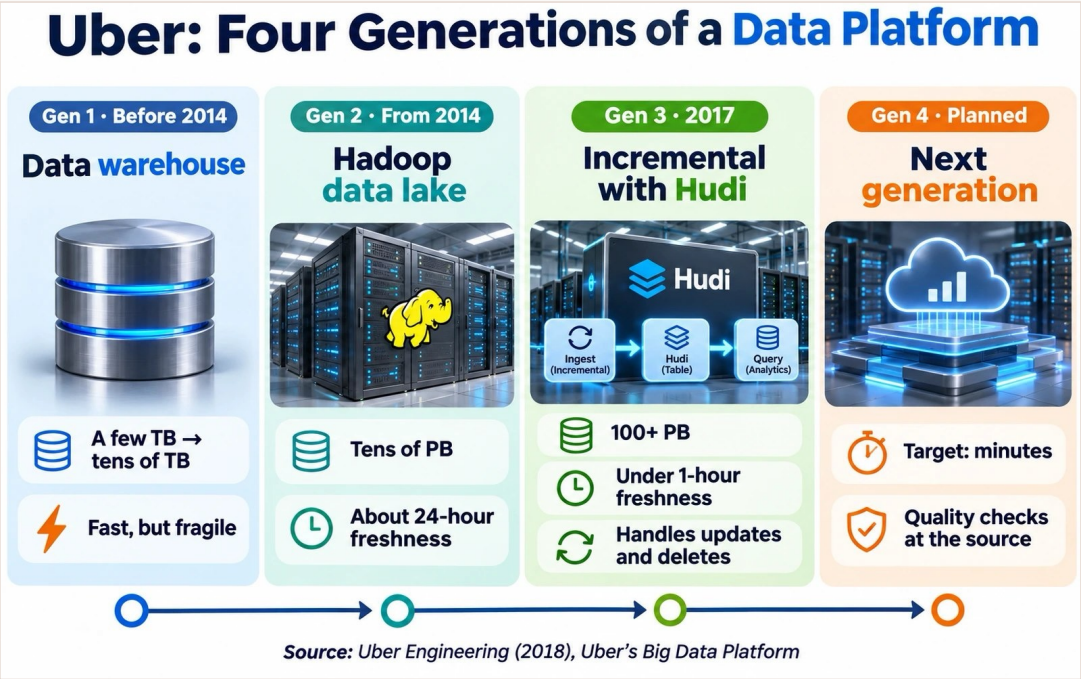


- Required: Uber, and Malaysia's COVID-19 open data
- Optional: PHE, Prime Video, Airflow, Kafka, Google Flu Trends
- Lens: era · lifecycle stage · undercurrent · trade-off

Source: Topic 3 web module, Case Studies

Required case 1

Uber relived the field's history

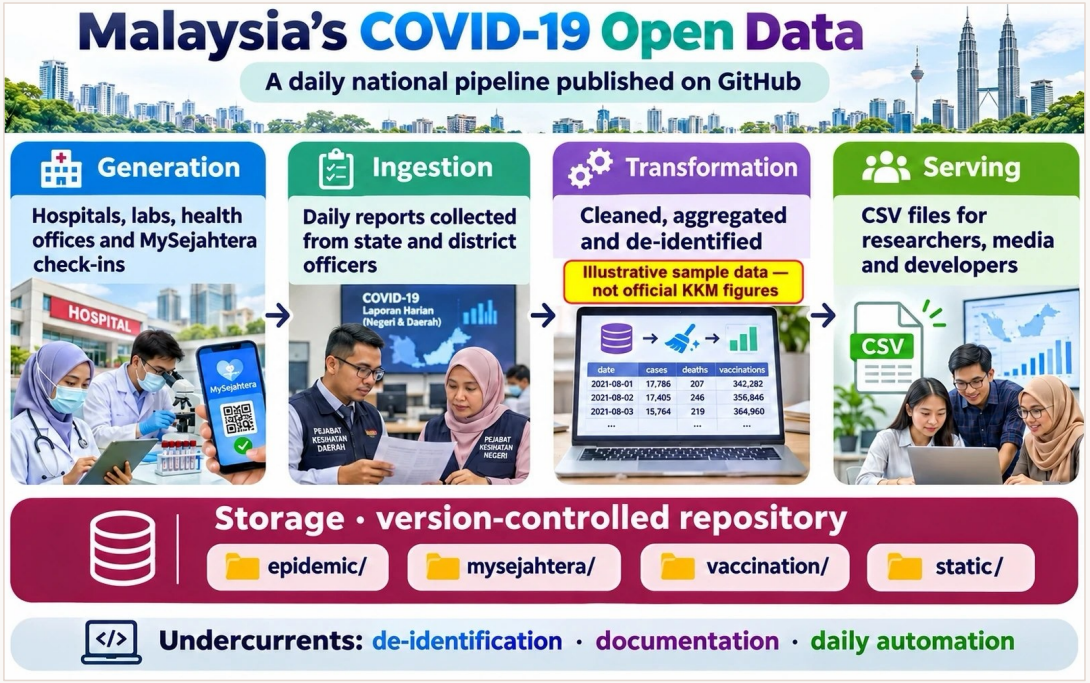


- Warehouse → Hadoop lake → incremental platform
- Freshness improved from 24+ hours to under 1 hour
- Each generation solved one problem and revealed the next

Source: Uber Engineering (2018), "Uber's Big Data Platform"

Required case 2 · Malaysia

A national pipeline on GitHub



- Daily CSV files published from mid-2021
- Folders: epidemic, mysejahtera, vaccination, static
- Undercurrents in focus: de-identification, documentation, automation

Source: KKM (2021); github.com/MoH-Malaysia/covid19-public

Optional cases · one per group for Part C

Five more stories to present

- Public Health England: what happens when a pipeline depends on a spreadsheet?
- Prime Video: when is the simpler architecture the better one?
- Airbnb and Airflow: why build an orchestrator—and give it away?
- LinkedIn and Kafka: how did the shift from batch to streaming begin?
- Google Flu Trends: why can big data still give the wrong answer?

Topic 3 web module, Optional case studies.

Case-study task · group · 10 marks

Two cases, one presentation

01 Part A · Uber: eras, lifecycle, trade-offs

02 Part B · KKM: lifecycle, undercurrents, privacy

03 Part C · Present one optional case in 3 minutes

04 Rubric: lifecycle 3 · evolution 2 · undercurrents 2

05 Rubric: presentation 2 · communication 1

06 Submit one PDF per group via UTM e-Learning

Worksheet and full rubric

Scan to copy the template

Key takeaways

Six ideas to remember

- Data engineering turns raw data into trustworthy information for others
- The field grew from warehousing to big data to lifecycle engineering
- Data engineering sits upstream: no foundation, no AI
- Five stages, six undercurrents: the map for this course
- Every design choice is a trade-off; SQL and Python matter
- Tools change quickly; fundamentals remain

Topic 3 web module, Reflection.

Before next class

Your next steps

01 Attempt the Topic 3 knowledge check

02 Write your reflection (200–300 words, ePortfolio)

03 Start the case-study task with your group

04 Read FDE Chapter 1, pages 3–21

05 Optional: watch the authors on the MAD Podcast

06 Next week: Topic 4 · data collection



Knowledge check

Scan to open the Topic 3 quiz



Foundations first.
Everything else follows.

Thank You

Assoc. Prof. Dr. Mohd Shahizan Othman

Universiti Teknologi Malaysia